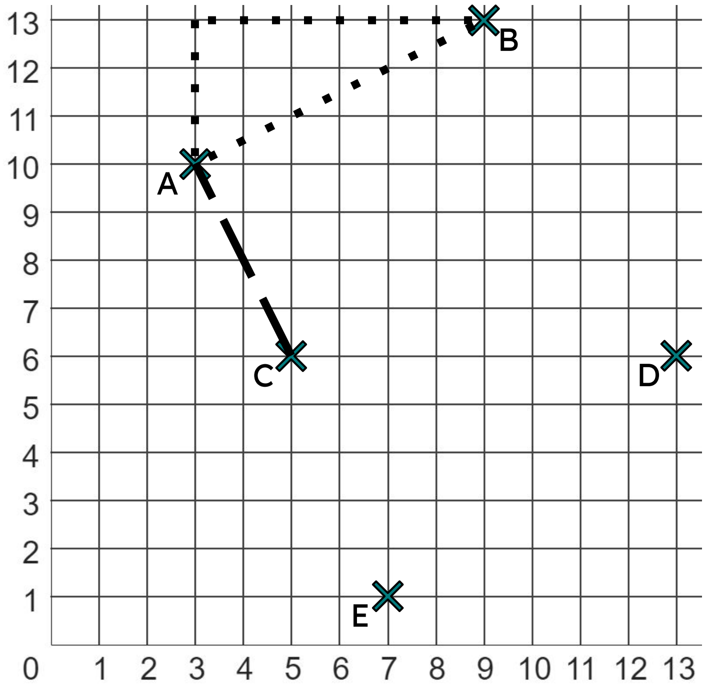
Name: ______________________

Worksheet

Problem solving with similarity and Pythagoras' theorem

Task A: Pythagoras' theorem with coordinates



1) Use Pythagoras' theorem to find the shortest distance (to 1 d.p) between points:

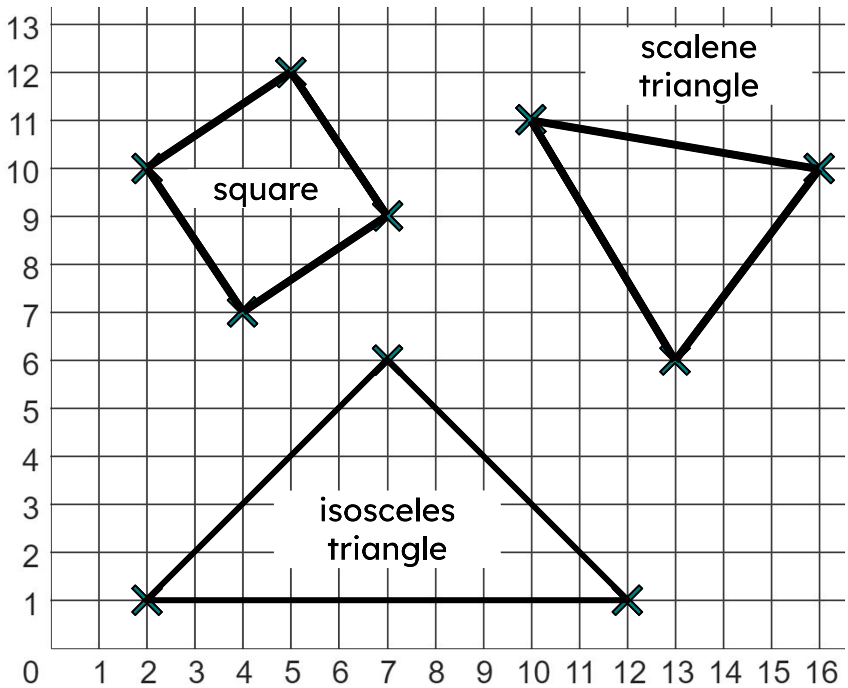
a) A and B

b) A and C

c) B and D

d) C and D

e) B and E

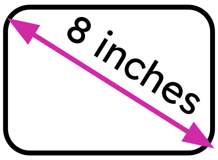


2) Calculate the perimeter of each shape (to 2 d.p.).

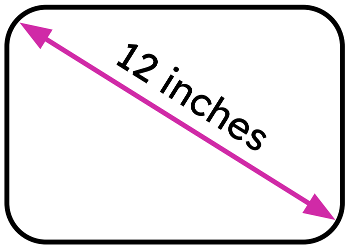
3) Write down three coordinates whose vertices make a triangle with a perimeter of between 20 and 22 units in length.
Show this is correct using Pythagoras' theorem.

Task B: Pythagoras' theorem with ratios

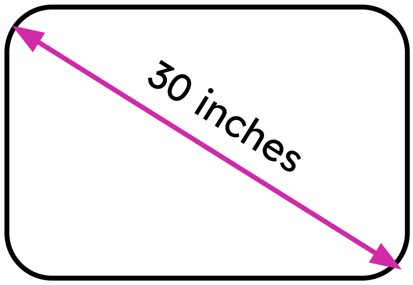
1) Each of these three widescreen TVs have width and height in the ratio 16 : 9.



screen A



screen B



screen C

a) Complete each sentence by choosing the appropriate word below.

Each of the TVs is __________ to the others.

This is because their ________ and ________ are in __________ to each other.

The __________ from screen A to B is 1.5.

proportion | scale factor | width | congruent | height | similar

b) By using Pythagoras' theorem, complete the ratio table for screen B (3 d.p.)

width	height	diagonal
16	9	
		12

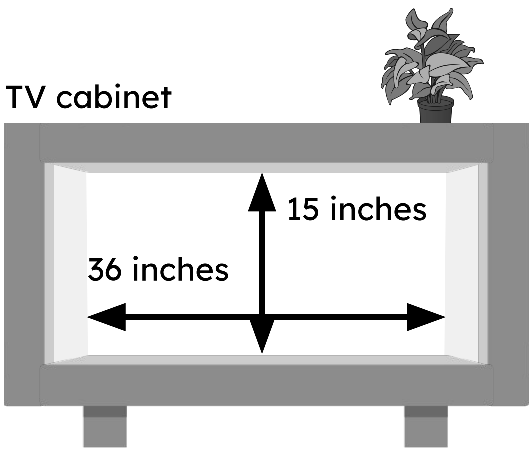
c) Hence, calculate the width of screens A and C (2 d.p.)

2) Izzy wants to buy this ultra widescreen TV, which has a width and height in the ratio 22 : 9.

The TV has a 35 inch screen, and a stand that is 2 inches high.

Will the TV fit inside her TV cabinet?

TV cabinet



width	height	diagonal

3) Jacob sees three TVs for sale, each having a 28 inch screen.

retro TV (4 : 3 aspect ratio)
widescreen TV (16 : 9 aspect ratio)
ultra widescreen TV (22 : 9 aspect ratio)

Jacob wants to buy the TV whose screen has the largest area.
Which TV should he buy?